

Multiple Choice

1. **Answer:** E

Options: A) 106.4 joule; B) 98.6 joule; C) 99.5 joule; D) 97.8 joule; E) 101.3 joule

Note: Correct option: E - 101.3 joule

2. **Answer:** A

Options: A) The function is continuous at (0, 0); B) The function is continuous along the line $y = x$; C) The function has a removable discontinuity at the origin; D) The function is continuous along the line $y = -x$; E) The function is continuous at all points

Note: Correct option: A - The function is continuous at (0, 0)

3. **Answer:** A

Options: A) inductance.; B) resistance.; C) impedance.; D) voltage.; E) capacitance.

Note: Correct option: A - inductance.

4. **Answer:** E

Options: A) $L\{t\} = (2 / s^3)$; B) $L\{t\} = (1 / s^3)$; C) $L\{t\} = s^2$; D) $L\{t\} = (1 / (2 s^2))$; E) $L\{t\} = (1 / s^2)$

Note: Correct option: E - $L\{t\} = (1 / s^2)$

5. **Answer:** A

Options: A) $\sinh x$; B) $e^x - e^{-x}$; C) $e^x + x$; D) $x^2 e^x$; E) $\cosh x$

Note: Correct option: A - $\sinh x$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $100 \cos 10^6 t$ '.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Voltage clamping device.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '1 5/8 inches'.

Long Form Response

11. **Answer:** 0.3 mm. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $D = f[(\sigma) / (g_c \rho d^2), \{\mu^2/(g\rho d^3)\}]$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key